

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 11: Finding Common Ground

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	11 — Finding Common Ground
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. HCF can never exceed the smaller number and uses the LOWEST powers of common primes, while LCM uses the HIGHEST powers of all primes and equals the product only for co-prime pairs. Do not write on this paper — mark answers on your answer sheet.

1. Find the HCF of 252, 378 and 630.

- | | |
|--------|---------|
| (A) 63 | (B) 42 |
| (C) 18 | (D) 126 |

2. Find the HCF of 96, 240 and 336.

- | | |
|--------|--------|
| (A) 48 | (B) 24 |
| (C) 16 | (D) 12 |

3. Find the LCM of 8, 9, 12 and 15.

- | | |
|---------|---------|
| (A) 180 | (B) 720 |
| (C) 120 | (D) 360 |

4. Find the LCM of 14, 21, 28 and 35.

- | | |
|---------|---------|
| (A) 210 | (B) 140 |
| (C) 420 | (D) 840 |

5. Find the HCF of 144, 180 and 192.

- | | |
|--------|--------|
| (A) 6 | (B) 12 |
| (C) 36 | (D) 24 |

6. Find the LCM of 6, 8, 10 and 15.

- (A) 60
(C) 30
- (B) 240
(D) 120

7. The number $588 = 2^2 \times 3 \times 7^2$. Using this, the HCF of 588 and $2^3 \times 3^2 \times 7$ is:

- (A) $2^3 \times 3^2 \times 7^2 = 3528$
(C) $2^2 \times 3^2 \times 7^2 = 1764$
- (B) $2^2 \times 3 \times 7 = 84$
(D) $2 \times 3 \times 7 = 42$

8. The number $588 = 2^2 \times 3 \times 7^2$. Using this, the LCM of 588 and $2^3 \times 3^2 \times 7$ is:

- (A) $2^3 \times 3^2 \times 7^2 = 3528$
(C) $2^3 \times 3^2 \times 7 = 504$
- (B) $2^2 \times 3 \times 7 = 84$
(D) $2^5 \times 3^3 \times 7^3$

9. If $1080 = 2^3 \times 3^3 \times 5$, how many of its factors are perfect squares (including 1)?

- (A) 4
(C) 8
- (B) 6
(D) 3

10. The total number of factors of $2^4 \times 3^2 \times 5$ is:

- (A) 11
(C) 30
- (B) 15
(D) 24

11. Two numbers are $2^3 \times 3^2 \times 5$ and $2^2 \times 3 \times 5^2$. The ratio LCM : HCF for them equals:

- (A) 6
(C) 15
- (B) 30
(D) 60

12. Two numbers have HCF 9 and LCM 360. If one number is 45, the other is:

- (A) 40
(C) 8
- (B) 72
(D) 81

13. The HCF of two numbers is 16 and their LCM is 240. How many such unordered pairs (a, b) exist?

- (A) 1
(C) 4
- (B) 3
(D) 2

14. Two numbers in the ratio 5 : 7 have an LCM of 700. The larger number is:

- (A) 100
(C) 35
- (B) 700
(D) 140

15. The LCM of two numbers is 1200 and their HCF is 20. If one number is 80, the other is:

- (A) 300
(C) 60
- (B) 240
(D) 1200

16. Two numbers have HCF 12. Their LCM is 72 times the HCF? No — their LCM is 504. The product of the two numbers is:

- (A) 516
(C) 6048
- (B) 42
(D) 6000

17. The HCF and LCM of two numbers are 6 and 90. The numbers are both multiples of 6 and lie between 10 and 50. The numbers are:

- (A) 6 and 90
(C) 12 and 45
- (B) 18 and 30
(D) 30 and 90

18. A gardener has 1024 marigolds and 768 roses. He arranges each kind into rows of equal, largest possible size with one kind per row. The number of flowers per row is:

- (A) 256
(C) 512
- (B) 128
(D) 192

19. Three ropes of lengths 7 m, 12.6 m and 16.8 m are cut into pieces of equal greatest possible length with none wasted. That length is:

- (A) 70 cm
(C) 35 cm
- (B) 140 cm
(D) 280 cm

20. A shopkeeper has 391 red beads, 425 blue beads and 527 green beads. He strings them so that each string holds beads of one colour only and every string has the same largest number of beads. That number is:

- (A) 23
(C) 13
- (B) 19
(D) 17

21. Square tiles of the largest possible whole-cm side pave a room 8 m 64 cm by 4 m 32 cm with no cutting. The side of each tile is:

- (A) 144 cm
(C) 216 cm
- (B) 432 cm
(D) 108 cm

22. From three drums holding 403 L, 434 L and 465 L of oil, the largest measuring can that can measure each an exact whole number of times has capacity:

- (A) 31 L
(C) 11 L
- (B) 13 L
(D) 7 L

23. Coloured cards numbering 504, 630 and 756 are bundled so each bundle has cards of one colour only and all bundles are equal and as large as possible. Each bundle contains:

- (A) 63
(C) 42
- (B) 126
(D) 18

24. Three runners finish a lap in 84, 105 and 140 seconds. Starting together, they are next all at the start together after:

- (A) 420 seconds
(C) 840 seconds
- (B) 210 seconds
(D) 329 seconds

25. Four bells toll at intervals of 5, 6, 8 and 9 seconds. If they toll together at the start, how many times do they toll together in 6 minutes (excluding the start)?

- (A) 2 (B) 3
(C) 6 (D) 1

26. Lights flash every 16, 24 and 40 seconds. After flashing together, the next simultaneous flash is after:

- (A) 120 seconds (B) 480 seconds
(C) 240 seconds (D) 80 seconds

27. Three electronic signs blink every 9, 15 and 25 minutes. They blink together at 12:00 noon. The next time all three blink together is:

- (A) 2:15 pm (B) 6:15 pm
(C) 3:45 pm (D) 1:15 pm

28. Cyclist A laps a track every 48 s, B every 60 s and C every 72 s. Starting together, the number of complete laps A has finished when they next all meet is:

- (A) 12 (B) 6
(C) 15 (D) 20

29. Two satellites orbit in 35 and 45 minutes. They align now. They will next align after how many minutes?

- (A) 1575 (B) 9
(C) 630 (D) 315

30. The product of two numbers is 3600 and their HCF is 12. Their LCM is:

- (A) 300 (B) 43200
(C) 30 (D) 120

31. For two numbers, $\text{HCF} = 15$ and $\text{LCM} = 4$ times the HCF wrongly stated; actually $\text{LCM} = 90$. The product of the numbers is:

- (A) 105 (B) 6
(C) 1350 (D) 1305

32. If the product of two numbers equals 8 times the square of their HCF, and their HCF is 9, their LCM is:

- (A) 72 (B) 648
(C) 81 (D) 9

33. Two numbers have an HCF of 23 and an LCM of 1449. The smaller of the two numbers (if the larger is 207) is:

- (A) 63 (B) 69
(C) 161 (D) 1449

34. For two numbers, the LCM is 12 times the HCF, and the HCF is 8. The product of the numbers is:

- | | |
|---------|---------|
| (A) 768 | (B) 104 |
| (C) 96 | (D) 12 |

35. The product of two co-prime numbers is 391. Their LCM is:

- | | |
|---------|--------|
| (A) 391 | (B) 1 |
| (C) 17 | (D) 23 |

36. The smallest number which, when divided by 24, 36 and 54, leaves a remainder of 7 in each case is:

- | | |
|---------|---------|
| (A) 216 | (B) 209 |
| (C) 223 | (D) 231 |

37. The largest number that divides 1356, 1868 and 2764 leaving the same remainder in each case is:

- | | |
|---------|---------|
| (A) 64 | (B) 256 |
| (C) 128 | (D) 32 |

38. A number leaves remainder 3 when divided by 5, remainder 3 when divided by 6, and remainder 3 when divided by 8. The smallest such number greater than 3 is:

- | | |
|---------|---------|
| (A) 120 | (B) 117 |
| (C) 123 | (D) 240 |

39. Find the greatest 3-digit number that leaves remainder 4 when divided by 6, 9 and 12.

- | | |
|---------|---------|
| (A) 972 | (B) 964 |
| (C) 940 | (D) 976 |

40. The smallest number which, when increased by 9, becomes exactly divisible by 15, 25 and 35 is:

- | | |
|---------|---------|
| (A) 516 | (B) 525 |
| (C) 534 | (D) 491 |

41. Which set of three numbers is pairwise co-prime (every pair has HCF 1)?

- | | |
|------------------|-------------------|
| (A) 6, 10 and 15 | (B) 8, 9 and 25 |
| (C) 4, 9 and 12 | (D) 12, 25 and 15 |

42. If a number is divisible by both 8 and 9, it must also be divisible by:

- | | |
|--------|---------|
| (A) 72 | (B) 17 |
| (C) 36 | (D) 144 |

43. Two co-prime numbers, each greater than 1, have a product of 1147. The two numbers are:

- | | |
|---------------|----------------|
| (A) 31 and 37 | (B) 1 and 1147 |
| (C) 29 and 39 | (D) 23 and 49 |

- 44.** For two numbers a and b , if $\text{HCF}(a, b) = 1$ then $\text{HCF}(a^2, b^2)$ equals:
 (A) a (B) b
 (C) ab (D) 1
- 45.** A number divisible by 11 and by 6 but NOT necessarily by 4 is always divisible by:
 (A) 44 (B) 66
 (C) 132 (D) 22
- 46.** If n is divisible by 12 and by 18, the smallest value n can take is:
 (A) 6 (B) 216
 (C) 36 (D) 72
- 47.** The number of common factors of 144 and 192 is:
 (A) 6 (B) 10
 (C) 8 (D) 12
- 48.** How many pairs of numbers (a, b) with $a < b$ have HCF 6 and LCM 90?
 (A) 1 (B) 3
 (C) 4 (D) 2
- 49.** How many factors of 360 are even?
 (A) 24 (B) 18
 (C) 12 (D) 6
- 50.** Among the numbers from 1 to 100, how many are common multiples of 4 and 6?
 (A) 16 (B) 8
 (C) 25 (D) 4
- 51.** The HCF of two numbers is 1 and one of them is 1. Which is true of their LCM and the other number n ?
 (A) $\text{LCM} = 1$ (B) $\text{LCM} = n$
 (C) $\text{LCM} = n^2$ (D) $\text{LCM} = n + 1$
- 52.** Two numbers have HCF 7 and their LCM is $7 \times 11 \times 13$. How many unordered pairs (a, b) with $a < b$ are possible?
 (A) 1 (B) 3
 (C) 4 (D) 2
- 53.** A rectangular field 1054 m by 952 m is divided into identical square plots of the greatest possible size. The number of plots along the longer side is:
 (A) 28 (B) 17
 (C) 31 (D) 34
- 54.** Bells P, Q and R toll every 6, 8 and 12 seconds. In one hour, how many times do all three toll together (counting the toll at the very start of the hour)?

(A) 150

(B) 75

(C) 151

(D) 301

55. The sum of two numbers is 528 and their HCF is 33. How many pairs of such numbers are possible?

(A) 6

(B) 4

(C) 2

(D) 8

56. Three bells start tolling together at time 0. They then toll at intervals of 4, 7 and 14 seconds. Counting from the start up to and including the next moment when all three toll together (but NOT counting the toll at time 0 itself), how many times does the FIRST bell (4-second interval) toll?

(A) 14

(B) 4

(C) 28

(D) 7

57. A two-digit number n satisfies $\text{HCF}(n, 24) = 8$ and $\text{LCM}(n, 24) = 96$. The number n is:

(A) 16

(B) 48

(C) 8

(D) 32

58. Two numbers are in the ratio 4 : 9 and their LCM is 180. Their HCF is:

(A) 5

(B) 20

(C) 45

(D) 36

59. A number leaves remainder 1 when divided by 2, 3, 4, 5 and 6. The smallest such number greater than 1 is:

(A) 60

(B) 31

(C) 121

(D) 61

60. The greatest length of a measuring rod that can measure exactly the lengths 3 m, 5 m 10 cm and 12 m 75 cm is:

(A) 5 cm

(B) 17 cm

(C) 15 cm

(D) 85 cm

Solutions — Mathematics (Class 7), Chapter 11: Finding Common Ground — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	B	31	C	41	B	51	B
2	A	12	B	22	A	32	A	42	A	52	D
3	D	13	D	23	B	33	C	43	A	53	C
4	C	14	D	24	A	34	A	44	D	54	C
5	B	15	A	25	D	35	A	45	B	55	B
6	D	16	C	26	C	36	C	46	C	56	D
7	B	17	B	27	C	37	C	47	B	57	D
8	A	18	A	28	C	38	C	48	D	58	A
9	A	19	B	29	D	39	D	49	B	59	D
10	C	20	D	30	A	40	A	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Factorise: $252 = 2^2 \times 3^2 \times 7$, $378 = 2 \times 3^3 \times 7$, $630 = 2 \times 3^2 \times 5 \times 7$. Common primes 2, 3, 7. Take the LOWEST power of each present in all three: $2^1, 3^2, 7^1$, giving $2 \times 9 \times 7 = 126$. 63 drops the factor 2; 42 wrongly uses 3^1 instead of 3^2 ; 18 drops the 7. HCF = 126.

2. (A) $96 = 2^5 \times 3$, $240 = 2^4 \times 3 \times 5$, $336 = 2^4 \times 3 \times 7$. Lowest powers of common primes: 2^4 and 3^1 , so HCF = $16 \times 3 = 48$. 24 uses 2^3 ; 16 forgets the shared 3; $12 = 2^2 \times 3$ takes too low a power of 2. HCF = 48.

3. (D) $8 = 2^3$, $9 = 3^2$, $12 = 2^2 \times 3$, $15 = 3 \times 5$. Highest powers: $2^3, 3^2, 5 = 8 \times 9 \times 5 = 360$. 180 misses one factor of 2 (8 needs 2^3); 720 doubles needlessly; 120 misses one factor of 3 (9 needs 3^2). LCM = 360.

4. (C) $14 = 2 \times 7$, $21 = 3 \times 7$, $28 = 2^2 \times 7$, $35 = 5 \times 7$. Highest powers: $2^2, 3, 5, 7 = 4 \times 3 \times 5 \times 7 = 420$. 210 uses only 2^1 ; 140 omits the factor 3; 840 doubles needlessly. LCM = 420.

- 5. (B)** $144 = 2^4 \times 3^2$, $180 = 2^2 \times 3^2 \times 5$, $192 = 2^6 \times 3$. Lowest common powers: 2^2 and 3^1 , so $\text{HCF} = 4 \times 3 = 12$. 6 takes 2^1 ; $36 = 2^2 \times 3^2$ wrongly keeps 3^2 (192 has only 3^1); $24 = 2^3 \times 3$ takes too high a power of 2. $\text{HCF} = 12$.
- 6. (D)** $6 = 2 \times 3$, $8 = 2^3$, $10 = 2 \times 5$, $15 = 3 \times 5$. Highest powers: 2^3 , 3, 5 = $8 \times 3 \times 5 = 120$. $60 = 2^2 \times 3 \times 5$ misses one factor of 2; 240 doubles; $30 = 2 \times 3 \times 5$ misses two factors of 2. $\text{LCM} = 120$.
- 7. (B)** HCF takes the lowest power of each shared prime. For 2: $\min(2,3)=2$; for 3: $\min(1,2)=1$; for 7: $\min(2,1)=1$. $\text{HCF} = 2^2 \times 3 \times 7 = 84$. The value 3528 is the LCM (highest powers); 1764 wrongly keeps 7^2 ; 42 wrongly uses 2^1 . $\text{HCF} = 84$.
- 8. (A)** LCM takes the highest power of each prime. For 2: $\max(2,3)=3$; for 3: $\max(1,2)=2$; for 7: $\max(2,1)=2$. $\text{LCM} = 2^3 \times 3^2 \times 7^2 = 3528$. 84 is the HCF (lowest powers); 504 wrongly uses 7^1 ; the last option adds the exponents instead of taking the maximum. $\text{LCM} = 3528$.
- 9. (A)** A factor $2^a \times 3^b \times 5^c$ is a perfect square when a, b, c are all even. a in $\{0,2\}$ (2 ways), b in $\{0,2\}$ (2 ways), c in $\{0\}$ (1 way, since exponent of 5 is only 1). Total = $2 \times 2 \times 1 = 4$; namely 1, 4, 9 and 36. Counting $c=1$ as allowed gives 6 (wrong, 5 is not squared); counting all factors loosely gives 8; forgetting 1 gives 3. Answer 4.
- 10. (C)** The number of factors is found by adding 1 to each exponent and multiplying: $(4+1)(2+1)(1+1) = 5 \times 3 \times 2 = 30$. Adding the exponents ($4+2+1=7$) and adding the primes is the common error giving 11 or similar; 15 forgets the +1 on one prime; $24 = 4 \times 3 \times 2$ forgets all the +1s. Answer 30.
- 11. (B)** HCF takes lowest powers: $2^2 \times 3 \times 5 = 60$. LCM takes highest powers: $2^3 \times 3^2 \times 5^2 = 1800$. Ratio = $1800/60 = 30$. In general $\text{LCM}/\text{HCF} = \text{product of } (\text{prime}^{\lvert \text{difference of exponents} \rvert}) = 2^{(3-2)} \times 3^{(2-1)} \times 5^{(2-1)} = 2 \times 3 \times 5 = 30$. Picking 6 uses only two primes; 15 drops the 2; 60 is the HCF itself. Ratio = 30.
- 12. (B)** $\text{HCF} \times \text{LCM} = \text{product of the numbers}$: $9 \times 360 = 45 \times \text{other}$, so $\text{other} = 3240/45 = 72$. Check $\text{HCF}(45,72)=9$, $\text{LCM}=360$. $40 = 360/9$ confuses LCM/HCF with the number; 8 is the cofactor $72/9$, not the number; 81 has LCM 405 with 45. Answer 72.
- 13. (D)** Write $a = 16m$, $b = 16n$ with $\text{HCF}(m,n) = 1$, and $\text{LCM} = 16mn = 240$ so $mn = 15$. Co-prime factorisations of 15: (1,15) and (3,5). That gives pairs (16,240) and (48,80). So 2 pairs. (3,5) and (1,15) are the only co-prime splits; (5,3) repeats. Counting ordered pairs gives 4; missing one split gives 1. Answer 2.
- 14. (D)** Let the numbers be $5k$ and $7k$. Since 5 and 7 are co-prime, $\text{HCF} = k$ and $\text{LCM} = 5 \times 7 \times k = 35k$. So $35k = 700$, $k = 20$. The numbers are 100 and 140; the larger is 140. 100 is the smaller; 700 is the LCM itself; $35 = 35 \times 1$ ignores k . Larger number = 140.

15. (A) $\text{HCF} \times \text{LCM} = \text{product}$: $20 \times 1200 = 24000 = 80 \times \text{other}$, so $\text{other} = 24000/80 = 300$. Check $\text{HCF}(80,300)=20$, $\text{LCM}=1200$. $240 = 1200/(60)$ is a guess; $60 = 1200/20$ confuses formula; 1200 is the LCM. Answer 300.

16. (C) Product of two numbers = $\text{HCF} \times \text{LCM} = 12 \times 504 = 6048$. $516 = 12 + 504$ adds instead of multiplying; $42 = 504/12$ is the LCM/HCF ratio; 6000 is a rounding slip. Product = 6048.

17. (B) Write $6m$ and $6n$ with co-prime m, n and $6mn = 90$ so $mn = 15 = 3 \times 5$. Co-prime pair $(3,5)$ gives 18 and 30, both between 10 and 50. $(1,15)$ gives 6 and 90, outside the range. 12 and 45 do not both share HCF 6 (45 is not even). 30 and 90 have HCF 30. Answer 18 and 30.

18. (A) Largest equal row dividing both is the HCF. $1024 = 2^{10}$, $768 = 2^8 \times 3$. Lowest common power of 2 is $2^8 = 256$ (the 3 is not shared). $128 = 2^7$ is common but not greatest; $512 = 2^9$ does not divide 768; $192 = 2^6 \times 3$ does not divide 1024. Answer 256.

19. (B) In cm: 700, 1260, 1680. $700 = 2^2 \times 5^2 \times 7$, $1260 = 2^2 \times 3^2 \times 5 \times 7$, $1680 = 2^4 \times 3 \times 5 \times 7$. Common primes 2, 5, 7 at lowest powers: $2^2 \times 5^1 \times 7^1 = 4 \times 5 \times 7 = 140$. So 140 cm. 70 uses 2^1 ; 35 drops the 4; $280 = 2^3 \times 5 \times 7$ takes too high a power of 2. Answer 140 cm.

20. (D) Need HCF of 391, 425, 527. $391 = 17 \times 23$, $425 = 5^2 \times 17$, $527 = 17 \times 31$. The only common prime is 17. HCF = 17. 23 divides only 391; 19 divides none; 13 divides none. Answer 17.

21. (B) In cm: 864 and 432. The largest tile is the HCF. Since 432 divides 864 exactly ($864 = 2 \times 432$), the HCF is 432. So tiles are 432 cm square: the room is 2 tiles by 1 tile. 144, 216 and 108 are smaller common divisors but not the greatest. Answer 432 cm.

22. (A) Need HCF of 403, 434, 465. $403 = 13 \times 31$, $434 = 2 \times 7 \times 31$, $465 = 3 \times 5 \times 31$. The only common prime is 31. HCF = 31. 13 divides only 403; 11 divides none; 7 divides only 434. Answer 31 L.

23. (B) HCF of 504, 630, 756. $504 = 2^3 \times 3^2 \times 7$, $630 = 2 \times 3^2 \times 5 \times 7$, $756 = 2^2 \times 3^3 \times 7$. Common primes 2, 3, 7 at lowest powers: $2^1 \times 3^2 \times 7 = 2 \times 9 \times 7 = 126$. 63 drops the 2; 42 uses 3^1 ; 18 drops the 7. Answer 126.

24. (A) Meeting time = LCM. $84 = 2^2 \times 3 \times 7$, $105 = 3 \times 5 \times 7$, $140 = 2^2 \times 5 \times 7$. Highest powers: $2^2, 3, 5, 7 = 4 \times 3 \times 5 \times 7 = 420$. 210 drops a factor of 2; 840 doubles; 329 is the sum. Answer 420 seconds.

25. (D) LCM of 5, 6, 8, 9: $5, 2 \times 3, 2^3, 3^2 \rightarrow 2^3 \times 3^2 \times 5 = 360$ seconds = 6 minutes. So they next toll together exactly at 360 s, the very end of 6 minutes — that is 1 time (excluding the start). 2 or 3 would need a shorter LCM. Answer 1.

26. (C) LCM of 16, 24, 40. $16 = 2^4$, $24 = 2^3 \times 3$, $40 = 2^3 \times 5$. Highest powers: $2^4 \times 3 \times 5 = 16 \times 15 = 240$. $120 = 2^3 \times 3 \times 5$ uses too low a power of 2; 480 doubles; $80 = 2^4 \times 5$ drops the 3. Answer 240 seconds.

27. (C) LCM of 9, 15, 25. $9 = 3^2$, $15 = 3 \times 5$, $25 = 5^2$. Highest powers: $3^2 \times 5^2 = 9 \times 25 = 225$ minutes = 3 h 45 min. $12:00 + 3:45 = 3:45$ pm. $2:15 = 135$ min misses a factor; $6:15 = 375$ min overshoots; $1:15 = 75$ min is too short. Answer 3:45 pm.

28. (C) Meeting time = LCM(48, 60, 72). $48 = 2^4 \times 3$, $60 = 2^2 \times 3 \times 5$, $72 = 2^3 \times 3^2$. Highest powers: $2^4 \times 3^2 \times 5 = 16 \times 9 \times 5 = 720$ s. A's laps = $720/48 = 15$. $12 = 720/60$ is B's laps; 6 is a slip; 20 confuses. A has done 15 laps.

29. (D) Next alignment = LCM(35, 45). $35 = 5 \times 7$, $45 = 3^2 \times 5$. Highest powers: $3^2 \times 5 \times 7 = 9 \times 5 \times 7 = 315$. 1575 = 35×45 wrongly takes the product (they are not co-prime, sharing factor 5); $9 = 45/5$; 630 doubles. Answer 315 minutes.

30. (A) HCF $\times$ LCM = product, so LCM = $3600/12 = 300$. Multiplying ($3600 \times 12 = 43200$) misapplies the rule; 30 and 120 do not satisfy $12 \times \text{LCM} = 3600$. LCM = 300.

31. (C) Product = HCF $\times$ LCM = $15 \times 90 = 1350$. $105 = 15 + 90$ adds; $6 = 90/15$ is the ratio; 1305 is a subtraction slip. Product = 1350.

32. (A) Product = $8 \times \text{HCF}^2 = 8 \times 81 = 648$. LCM = product/HCF = $648/9 = 72$. Equivalently LCM = $8 \times \text{HCF} = 72$ since product = HCF $\times$ LCM = $8 \times \text{HCF}^2$. 648 is the product, not LCM; $81 = \text{HCF}^2$; 9 is the HCF. LCM = 72.

33. (C) Product = HCF $\times$ LCM = $23 \times 1449 = 33327 = 207 \times \text{smaller}$, so smaller = $33327/207 = 161$. Check HCF(161,207): $161 = 7 \times 23$, $207 = 9 \times 23$, HCF = 23, LCM = $7 \times 9 \times 23 = 1449$. 63 and 69 do not pair correctly; 1449 is the LCM. Smaller = 161.

34. (A) LCM = $12 \times \text{HCF} = 12 \times 8 = 96$. Product = HCF $\times$ LCM = $8 \times 96 = 768$. $104 = 8 + 96$ adds; 96 is the LCM only; 12 is the multiple factor. Product = 768.

35. (A) For co-prime numbers the HCF is 1, so LCM = product/HCF = $391/1 = 391$. (Here $391 = 17 \times 23$, two co-primes.) 1 is the HCF; 17 and 23 are the individual numbers. LCM = 391.

36. (C) First find LCM(24, 36, 54). $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, $54 = 2 \times 3^3$. Highest powers: $2^3 \times 3^3 = 8 \times 27 = 216$. The smallest such number is LCM + remainder = $216 + 7 = 223$. 216 forgets to add 7; $209 = 216 - 7$ subtracts; $231 = 224 + 7$ uses a wrong LCM. Answer 223.

37. (C) When the same remainder is left, the number divides the differences. $1868 - 1356 = 512$, $2764 - 1868 = 896$, $2764 - 1356 = 1408$. HCF of 512 (2^9), 896 ($2^7 \times 7$), 1408 (2^7

$\times 11) = 2^7 = 128$. $64 = 2^6$ is common but not greatest; $256 = 2^8$ does not divide 896; $32 = 2^5$ is too small. Answer 128.

38. (C) The number minus 3 is divisible by 5, 6 and 8, so it is a multiple of $\text{LCM}(5,6,8)$. $6 = 2 \times 3$, $8 = 2^3$, $5 = 5$: $\text{LCM} = 2^3 \times 3 \times 5 = 120$. Smallest such number above 3 is $120 + 3 = 123$. 120 forgets the +3; $117 = 120 - 3$; 240 is the next multiple. Answer 123.

39. (D) The number minus 4 must be a multiple of $\text{LCM}(6,9,12)$. $6 = 2 \times 3$, $9 = 3^2$, $12 = 2^2 \times 3$: $\text{LCM} = 2^2 \times 3^2 = 36$. Largest 3-digit multiple of 36 is 972 (36×27). Add 4: 976, still 3-digit. 972 forgets the +4; 964 uses a wrong multiple; 940 is not of the right form. Answer 976.

40. (A) We need $\text{LCM}(15, 25, 35)$ then subtract 9. $15 = 3 \times 5$, $25 = 5^2$, $35 = 5 \times 7$: $\text{LCM} = 3 \times 5^2 \times 7 = 525$. The number is $525 - 9 = 516$. 525 forgets the -9; $534 = 525 + 9$ adds instead; 491 uses a wrong LCM. Answer 516.

41. (B) Pairwise co-prime means each pair shares no prime factor. $8 = 2^3$, $9 = 3^2$, $25 = 5^2$ share no primes pairwise — co-prime. In 6,10,15 each pair shares a factor (2, 3 or 5). In 4,9,12: 4 and 12 share 4. In 12,25,15: 12 and 15 share 3. Answer 8, 9 and 25.

42. (A) Numbers divisible by both 8 and 9 are exactly the multiples of $\text{LCM}(8,9)$. Since 8 and 9 are co-prime, $\text{LCM} = 72$. So such a number is always divisible by 72. $17 = 8 + 9$ adds; $36 = \text{LCM}(4,9)$ is too small (e.g. 72 is divisible by 8 and 9 but 36 only divides 72, while 72 always divides — yet not every multiple of 8 and 9, like 72 itself, fails 36? 72 is divisible by 36, but the guaranteed divisor is 72); 144 is not guaranteed (72 is divisible by 8 and 9 but not 144). Answer 72.

43. (A) Since the numbers are co-prime, their product equals their LCM and each is built from distinct primes. Factor $1147 = 31 \times 37$, both prime, hence co-prime and both above 1. 1 and 1147 fails the 'greater than 1' condition; $29 \times 39 = 1131$, not 1147; $23 \times 49 = 1127$, not 1147. Answer 31 and 37.

44. (D) If a and b share no prime factor, then a^2 and b^2 also share no prime factor, so their HCF is 1. Squaring cannot create a new common prime. The options a, b, ab assume a shared factor that does not exist. Answer 1.

45. (B) Divisible by 11 and 6 means divisible by $\text{LCM}(11,6) = 66$ (since 11 and 6 are co-prime). It need not be divisible by 4, so we cannot claim $44 = 11 \times 4$ or $132 = 11 \times 12$. $22 = 11 \times 2$ is guaranteed but is not the largest guaranteed divisor; the question asks for what it is always divisible by from the list, and 66 is the LCM. Answer 66.

46. (C) n must be a common multiple of 12 and 18; the smallest is $\text{LCM}(12,18)$. $12 = 2^2 \times 3$, $18 = 2 \times 3^2$: $\text{LCM} = 2^2 \times 3^2 = 36$. 6 is the HCF (a divisor, not a multiple); $216 = 12 \times 18$ takes the product; 72 is a larger common multiple. Smallest n = 36.

47. (B) Common factors are factors of $\text{HCF}(144, 192)$. $144 = 2^4 \times 3^2$, $192 = 2^6 \times 3$: $\text{HCF} = 2^4 \times 3 = 48$. Number of factors of 48 = $(4+1)(1+1) = 5 \times 2 = 10$. 6 forgets a +1; 8 miscounts; 12 uses a wrong HCF. Answer 10.

48. (D) Let $a = 6m$, $b = 6n$, $\text{HCF}(m, n) = 1$, and $\text{LCM} = 6mn = 90$ so $mn = 15$. Co-prime splits of 15: (1,15) and (3,5), giving (6,90) and (18,30). So 2 unordered pairs with $a < b$. Counting ordered gives 4; missing a split gives 1. Answer 2.

49. (B) $360 = 2^3 \times 3^2 \times 5$. Total factors = $(3+1)(2+1)(1+1) = 24$. Odd factors come from 2^0 : $(2+1)(1+1) = 6$. Even factors = total - odd = $24 - 6 = 18$. 24 counts all; 12 halves wrongly; 6 is the odd count. Answer 18.

50. (B) Common multiples of 4 and 6 are multiples of $\text{LCM}(4, 6) = 12$. Multiples of 12 up to 100: 12, 24, 36, 48, 60, 72, 84, 96 — that is 8 numbers. 16 counts multiples of 6; 25 counts multiples of 4; 4 undercounts. Answer 8.

51. (B) Since 1 divides everything, $\text{LCM}(1, n) = n$ and $\text{HCF}(1, n) = 1$. So the LCM equals the other number n . $\text{LCM} = 1$ only if both were 1; n^2 and $n+1$ misapply the formula. Answer $\text{LCM} = n$.

52. (D) Write $a = 7m$, $b = 7n$ with co-prime m, n and $mn = 11 \times 13 = 143$. Co-prime splits of $143 = 11 \times 13$: (1,143) and (11,13). That gives pairs (7,1001) and (77,91). So 2 pairs. In general the count is $2^{(k-1)}$ where k = number of distinct primes in LCM/HCF ; here $k = 2$, so 2. Answer 2.

53. (C) Plot side = $\text{HCF}(1054, 952)$. $1054 = 2 \times 17 \times 31$, $952 = 2^3 \times 7 \times 17$. Common: $2 \times 17 = 34$. So plots are 34 m square. Along the 1054 m side: $1054/34 = 31$ plots. $28 = 952/34$ is along the shorter side; 17 and 34 confuse the side length with the count. Answer 31.

54. (C) All toll together every $\text{LCM}(6, 8, 12) = 24$ seconds ($6 = 2 \times 3$, $8 = 2^3$, $12 = 2^2 \times 3 \rightarrow 2^3 \times 3 = 24$). In 3600 seconds the simultaneous tolls occur at 0, 24, 48, ... up to 3600. Count = $3600/24 + 1 = 150 + 1 = 151$. 150 forgets the start toll; 75 halves; 301 double counts. Answer 151.

55. (B) Write the numbers as $33a$ and $33b$ with co-prime a, b . Then $33(a + b) = 528$, so $a + b = 16$. Co-prime pairs summing to 16 with $a < b$: (1,15), (3,13), (5,11), (7,9). That is 4 pairs. Pairs like (2,14), (4,12) share a factor; (8,8) is not distinct/co-prime. Answer 4.

56. (D) All three toll together again after $\text{LCM}(4, 7, 14)$ seconds. Since $14 = 2 \times 7$ and $4 = 2 \times 2$, $\text{LCM}(4, 7, 14) = 28$ s. The first bell tolls every 4 s, at $t = 4, 8, 12, 16, 20, 24, 28$. Excluding the toll at $t = 0$ and including the simultaneous toll at $t = 28$, that is $28 \div 4 = 7$ tolls. Answering 14 uses the wrong interval (the 2-second half-step); 4 just repeats an interval; 28 is the cycle length in seconds, not a count.

57. (D) Using $\text{HCF} \times \text{LCM} = \text{product}$: $8 \times 96 = 768 = n \times 24$, so $n = 768/24 = 32$. Check: $\text{HCF}(32,24)=8$, $\text{LCM}(32,24)=96$, and 32 is two-digit. 16 has LCM 48 with 24; 48 has HCF 24 with 24; 8 is the HCF itself. Answer 32.

58. (A) Let the numbers be $4k$ and $9k$. Since 4 and 9 are co-prime, the HCF is k and the LCM is $4 \times 9 \times k = 36k$. So $36k = 180$ gives $k = 5$, which is the HCF. The numbers are 20 and 45. 20 and 45 are the numbers themselves, not the HCF; $36 = 36 \times 1$ ignores k . HCF = 5.

59. (D) The number minus 1 must be divisible by 2, 3, 4, 5 and 6, so it is a multiple of $\text{LCM}(2,3,4,5,6)$. $4 = 2^2$, $6 = 2 \times 3$, $5 = 5$, $3 = 3$: $\text{LCM} = 2^2 \times 3 \times 5 = 60$. Smallest such number above 1 is $60 + 1 = 61$. 60 forgets the +1; 31 uses a wrong LCM; 121 is the next multiple plus 1. Answer 61.

60. (C) In cm: 300, 510, 1275. HCF: $300 = 2^2 \times 3 \times 5^2$, $510 = 2 \times 3 \times 5 \times 17$, $1275 = 3 \times 5^2 \times 17$. Common primes 3 and 5 at lowest powers: $3 \times 5 = 15$. So 15 cm. 5 takes only the 5; 17 divides 510 and 1275 but not 300; $85 = 5 \times 17$ does not divide 300. Answer 15 cm.